



BUILD WITH BHARAT 2.0 | NIT DELHI | ONLINE ROUND

URJASETU

Keep the solar. Prove the extra benefit.

Green Energy Management | Sustainability

Supported retrofit monitoring, useful task scheduling and transparent savings.



STAGE: PROTOTYPE READY (TEAM-REPORTED) | REAL-HOME VALIDATION NEXT

TEAM: Code Warriar 2.0

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PROBLEM STATEMENT

Solar adoption is growing. Useful decisions still need context.

FIRST USER

A rooftop-solar household or small building with a flexible water-pump task.

THE DECISION

When should the pump run, can it finish before the deadline, and does changing its timing actually reduce net cost?

50.06 lakh+

PM Surya Ghar beneficiary households.
Reported 4 Aug 2026; 14.8 GW
commissioned under the scheme.

Adoption context, not a claim that every
household needs UrjaSetu.

CHECK BENEFIT BEFORE ADDING HARDWARE

Configure the actual tariff, export credits and device compatibility. With flat tariffs and fully usable 1:1 net-metering credits, timing-only shifts may save little or nothing. Exported solar is not automatically wasted.



SOLUTION

Keep existing panels and inverter. Add only what is missing.

ROUTE A: USE AVAILABLE DATA

Owner-approved, model-supported API / Modbus telemetry. Fronius Solar API is one documented read-data route.

ROUTE B: METER THE REQUIRED CIRCUITS

A qualified installer adds suitable AC metering / CTs in the distribution board. Gateway communicates over LAN / Wi-Fi.

CONTROL IS A SEPARATE PATH

Use a supported actuator / rated contactor on the flexible-load circuit. A read API alone cannot switch a pump.

No utility-meter seal tampering. No DIY mains switching. Keep manual override and load-specific protection.

TRACK

Generation, load and grid import / export.

Sources: Fronius Solar API (read telemetry)

OPTIMIZE

Choose a feasible task slot; obtain approval.

CARBON

Indicative import footprint; show factor/year.

Shelly technical guide (metering / contactor distinction)

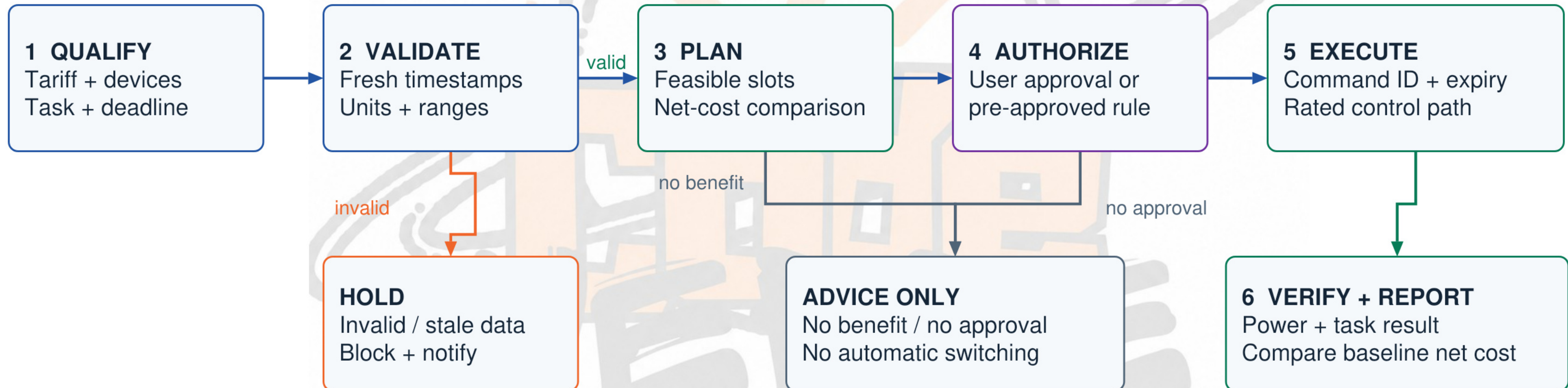
GRID

Tariff-aware planning; utility programs need authorization.



FLOW OF SOLUTION

Proposed control loop: no blind command, no unverified receipt.



Delivery acknowledgment is not physical execution. Missing feedback -> unverified result + alert.

Safety requirements: protect critical loads; time-bound commands; no unsafe replay after reconnect; manual fallback.

Reference: Espressif ESP-MQTT. Safety states shown are design requirements, not confirmed test results.



TECH STACK

A small, rules-first system with a defined upgrade path.

PROPOSED ARCHITECTURE FROM THE PROJECT PLAN

EDGE / SENSING	ESP32 + INA219 + low-voltage DC load	Demo reference; DC sensor is not a mains energy meter.
TRANSPORT	MQTT / HTTPS	Require TLS, per-device credentials and bounded commands.
BACKEND	Python FastAPI	Validate readings, authorize actions and record outcomes.
STORAGE / UI	SQLite + React	Telemetry, tasks, receipts; PostgreSQL is a later scale path.
OPTIMIZATION	Rules + feasible-slot enumeration	Use solar, actual tariff, runtime, deadline and load limits.

AI NEXT: train a solar/load forecast only with suitable data; compare held-out error against a naive forecast.

LLM explanations must not bypass rules. Mark every input as measured, simulated or forecast; disclose implemented modules.

Technical reference: Espressif ESP-MQTT. Library support does not prove correct prototype configuration.



UNIQUE SELLING PROPOSITION

RETROFIT > TASK > PROOF

RETROFIT

Check compatibility and financial benefit first.
Reuse supported interfaces; avoid unnecessary hardware.

TASK

Complete useful work, not just an ON/OFF toggle.
Example: run a pump for 30 minutes before 4 PM.

PROOF

Separate existing solar benefit from the optimizer's additional benefit. Compare with a simple fixed timer.

SAVINGS RECEIPT: WORKED EXAMPLE

Illustration only - not measured prototype results

Task energy: 0.5 kWh
Assumed import value: INR 8 / kWh
Assumed foregone export value: INR 3 / kWh

$$0.5 \times (8 - 3) = \text{INR } 2.50$$

Incremental benefit = baseline net cost - optimized net cost.
Assumes available surplus PV and separately valued exports.
Flat tariff + fully usable 1:1 credits can make this benefit zero.

Differentiation is the focused workflow, not a claim that competing schedulers do not exist.

Economic boundary: BSES net-metering accounting. Actual customer tariff and settlement rules take precedence.



FEASIBILITY & COMPETITORS

REALISTIC DEPLOYMENT

Prototype: ready (team-reported).

Pilot: verify one device route, safety and tariff.

Scale: compatibility and field reliability unproven.

EXISTING CAPABILITIES

SolarEdge ONE: AI planning; supported ecosystem plus selected third-party devices.

Schneider Wiser: predictive home-load scheduling and user override on supported hardware/markets.

Shelly: local metering, integrations and configurable control; suitable external actuator required.

AFFORDABILITY GATE

Installed cost / monthly net benefit = payback

Illustration: INR 100 net benefit / month x 24 months
= INR 2,400 maximum installed cost for this target.

Target scenario, not a hardware quote or achieved saving.
Include installation, recurring fees and maintenance.

COMPETE ON FIT + EVIDENCE

A narrow pump use case, supported retrofit route and auditable cost comparison. Superiority must be tested.

Commercial hypothesis: installer-assisted setup + optional service. Price only after BOM and real-home benefit validation.



RESEARCH & VALIDATION

PROTOTYPE STATUS

Ready, as reported by the team. This deck does not include a demo URL, screenshot, BOM or test log.

VALIDATION PROTOCOL

Run usual schedule, fixed solar-hours timer and UrjaSetu on identical load, solar and tariff scenarios.

REPORT

Task completion, grid import, net cost, blocked unsafe commands and actual execution feedback. Label measured versus simulated data.

CARBON: imported kWh x selected CEA factor; show factor year and method.

Indicative import footprint, not certified carbon credits or proven system-wide avoided emissions.

PRIMARY RESEARCH

MNRE / PIB - PM Surya Ghar, 4 Aug 2026

Ministry of Power - ToD, 2 Aug 2023

BSES - net-metering accounting summary

SolarEdge ONE - product capabilities

Schneider - Wiser launch, 6 Sep 2024

Fronius - Solar API / telemetry

Shelly - Pro EM-50 technical guide

Espressif - ESP-MQTT guide

CEA - CO2 Baseline Database v21

Research accessed 31 Aug 2026. Vendor capabilities are not UrjaSetu test results. Public sources do not validate prototype performance.